

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4302

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`

D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

Snippet 4: XML Syntax

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`
- D. `<note to="User"></note>`

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I THIRUMALAPUDI PRANAVI (192411222) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T.PRANAVI

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

WEB DEVELOPMENT STANDARDS IMPLEMENTATION

Scenario

A startup is developing a modern web application and faces several issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols. To ensure that the application follows proper web development standards, the correct technologies and syntax must be identified and applied.

The main requirements are:

1. Embed a promotional video on the homepage.
2. Securely submit registration form data to the server.
3. Use semantic HTML for the navigation menu.
4. Correct syntax errors in XML configuration files.
5. Use the appropriate protocol for communication between the client and server.

1. HTML Video Embedding

Given Options

A) `<media src="video.mp4"></media>`

B) `<video src="video.mp4" controls></video>`

C) `<movie src="video.mp4"></movie>`

D) `<vid src="video.mp4"></vid>`

Correct Answer

B) `<video src="video.mp4" controls></video>`

Explanation

HTML5 provides the `<video>` element specifically for embedding video content directly into a webpage. The `controls` attribute provides standard controls such as play, pause, volume, and video progress.

Code

```
<video controls width="600">
```

```
  <source src="video.mp4" type="video/mp4">
```

```
  Your browser does not support the video element.
```

```
</video>
```

Justification

The <video> element is a standard HTML5 semantic media element. It allows developers to embed videos without requiring external plugins.

The other options are incorrect because:

- <media> is not a standard HTML video element.
- <movie> is not a standard HTML element.
- <vid> is not a valid HTML5 video element.

2. HTTP Method for Form Submission

Given Options

- A) GET
- B) POST
- C) PUT
- D) DELETE

Correct Answer

B) POST

Explanation

The **POST** method is appropriate for submitting registration data because the information is sent in the HTTP request body rather than being included in the URL.

For example:

```
<form action="/register" method="POST">
  <input type="text" name="username">
  <input type="email" name="email">
  <input type="password" name="password">
  <button type="submit">Register</button>
</form>
```

Why POST is Suitable

- Data is sent in the request body.
- Sensitive information is not exposed in the URL.

- It is suitable for creating or submitting data to a server.
- It supports larger amounts of form data than a URL-based GET request.

Important: POST alone does **not encrypt** the data. For secure transmission, the application should use HTTPS.

Why the Other Methods Are Less Suitable

- **GET:** Mainly used to retrieve data; form data appears in the URL.
- **PUT:** Generally used to create or completely update a resource.
- **DELETE:** Used to remove a resource.

3. Semantic Element for Navigation

Given Options

- A) <div>
- B) <nav>
- C) <section>
- D) <header>

Correct Answer

- B) <nav>

Explanation

The HTML5 <nav> element is specifically designed to contain a group of important navigation links.

Example

```
<nav>
  <a href="index.html">Home</a>
  <a href="about.html">About</a>
  <a href="services.html">Services</a>
  <a href="contact.html">Contact</a>
</nav>
```

Justification

Using <nav> clearly communicates the purpose of the content to browsers, search engines, and assistive technologies such as screen readers.

It improves:

- Accessibility
- Code readability
- Semantic structure
- Search engine understanding

The <div> element is a generic container, while <section> is used for a thematic section of content. <header> represents introductory content and is not specifically intended for navigation.

4. Correct XML Syntax

Given Options

A)

```
<note>  
  <to>User</to>  
  <from>Admin</from>
```

B)

```
<note>  
  <to>User</to>  
  <from>Admin</from>  
</note>
```

C)

```
<note>  
  <to>User</from>  
</note>
```

D)

```
<note to="User"></note>
```

Correct Answer

B)

```
<note>  
  <to>User</to>  
  <from>Admin</from>  
</note>
```

Explanation

XML requires elements to be properly opened and closed, and the opening and closing tags must match.

Errors in Other Options

Option A:

The `<note>` element is opened but not closed.

Option C:

The opening tag is `<to>` but it is incorrectly closed using `</from>`. XML tags must match.

Option D:

The closing `>` is missing from the `</note>` tag.

Important XML Rules

1. Every opened element must be closed.
2. Opening and closing tags must match.
3. XML elements must be properly nested.
4. XML is case-sensitive.
5. Attribute values must be enclosed in quotation marks.

Therefore, **Option B follows correct XML syntax.**

5. Protocol for Web Communication

Given Options

- A) FTP
- B) HTTP
- C) SMTP
- D) SNMP

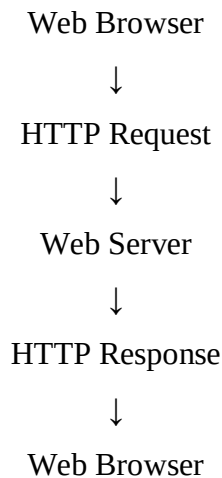
Correct Answer

B) HTTP

Explanation

HTTP (Hypertext Transfer Protocol) is the standard application-layer protocol used for communication between web browsers and web servers.

The basic communication process is:



For example, when a user opens a webpage, the browser sends an HTTP request to the server. The server processes the request and returns an HTTP response containing resources such as HTML, CSS, JavaScript, images, or other data.

Other Protocols

- **FTP:** Used mainly for transferring files.
- **SMTP:** Used for sending email.
- **SNMP:** Used for network management and monitoring.

Security Consideration

Modern websites should use **HTTPS** rather than plain HTTP. HTTPS uses encryption through TLS to protect data transmitted between the browser and server.

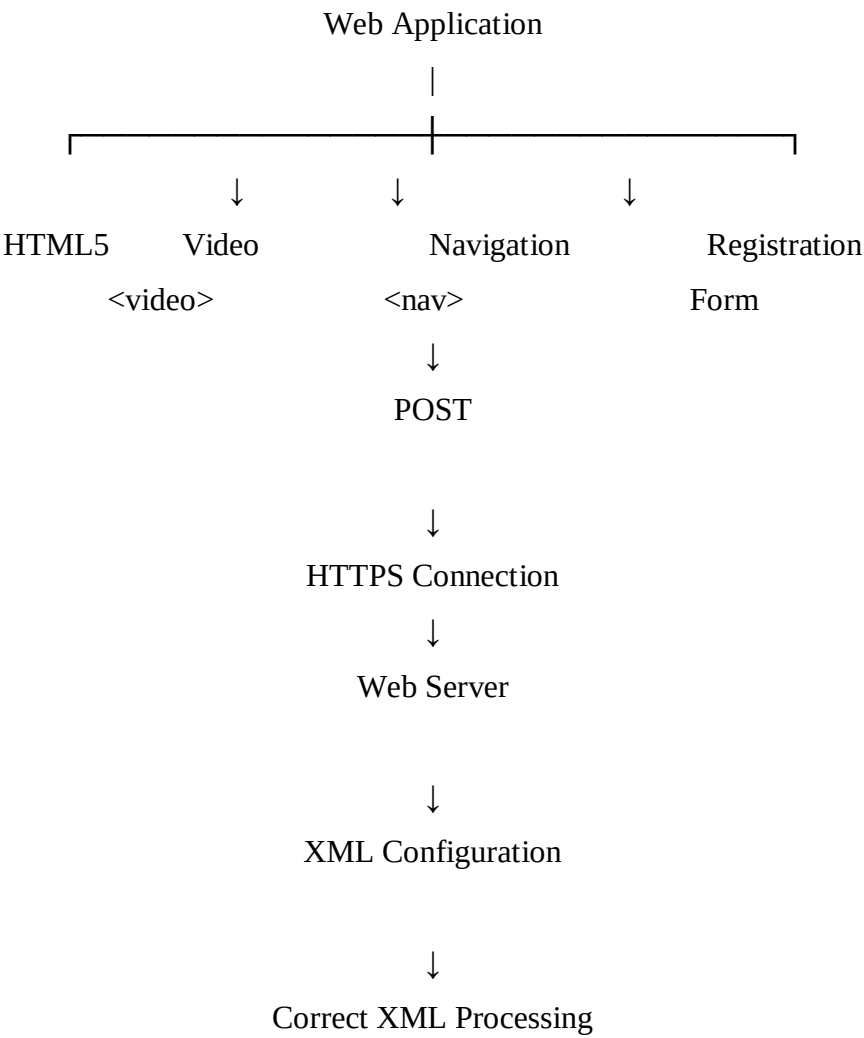
Overall Solution

The startup can solve the identified web development issues by applying standard web technologies correctly:

Requirement	Appropriate Solution
Embed promotional video	HTML5 <video>
Submit registration data	HTTP POST
Create navigation menu	Semantic <nav>
Correct XML configuration	Proper opening, closing, and nesting of tags

Web communication	HTTP/HTTPS
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Integrated Web Application Flow



Conclusion

Following web development standards is essential for building a reliable, accessible, and secure web application. The HTML5 `<video>` element should be used for video embedding, while the POST method is appropriate for submitting registration data. The semantic `<nav>` element should be used for navigation links, and XML files must follow proper syntax and nesting rules. HTTP is the fundamental protocol for

web communication, while HTTPS should be preferred when secure communication is required.

By applying these standards correctly, the startup can improve the application's functionality, security, accessibility, maintainability, and overall user experience.